

Linyi (Tiger) Hou

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EDUCATION

Ph.D. in Aerospace Engineering, University of Illinois Urbana-Champaign
B.S. in AE, Minor in CompE, University of Illinois Urbana-Champaign

Exp. May 2025
Aug 2017 - May 2021

SKILLS

Programming Languages	Python, C++, MATLAB/Simulink
Libraries	numpy astropy scipy pybind11, Eigen OpenMP
Tools	Git, GitHub, GMAT, Slurm
Operating Systems	Linux, MacOS, Windows, FreeRTOS

HIGHLIGHTS FOR STATE ESTIMATION AND MODELING ENGINEER

- ❖ Proficient in using C++/Python-driven simulations to evaluate the performance of navigation algorithms.
- ❖ Experienced in high-precision timing and positioning in the context of X-ray pulsar navigation.
- ❖ Experienced in the modeling and simulation of sensors (navigation cameras and star trackers), system dynamics (orbital mechanics), and the space environment (spherical harmonics).
- ❖ Designed, implemented, and analyzed novel PNT techniques culminating in three journal publications.

EXPERIENCE

Graduate Researcher, Astrodynamics and Planetary Exploration Group (APEX)

Interstellar Navigation and Star Identification

Sep 2024 - present

- ❖ Developing star identification techniques for interstellar travel at relativistic speeds by combining traditional algorithms with clustering based on stellar classification.

Celestial Navigation to Supplement DSN Tracking

Jul 2024 - Mar 2025

- ❖ Collaborated with four colleagues to assess feasibility of using celestial navigation to meet cruise phase uncertainty requirements in heliocentric and cislunar space.
- ❖ Designed sensor tasking/scheduling algorithms with novel considerations for covariance propagation and visibility constraints over the time domain.
- ❖ Implemented spherical harmonics for an open source orbit propagator in C++.

Lost in Space and Time Navigation using Variable Stars

Nov 2022 - Mar 2025

- ❖ Developed an autonomous optical navigation technique based on light curve analysis of variable stars which does not require prior knowledge of position or time.
- ❖ Simulated star tracker observations and performed Monte Carlo analysis to assess the feasibility of reestablishing Earth pointing via the proposed method.

Graduate Researcher, Aerospace Mission Analysis Laboratory

X-ray Pulsar Navigation (XNAV)

Jun 2021 - Dec 2023

- ❖ Developed algorithms to resolve spacecraft position and time from pulsar timing data.
- ❖ Created a photon time of arrival simulation tool in Python and validated results against the pulsar timing package [PINT](#).

Velocity-based Initial Orbit Determination

Aug 2019 - Apr 2024

- ❖ Developed a novel initial orbit determination (IOD) technique using sequential range-rate measurements to distant stars.
- ❖ Assessed the performance of velocity IOD methods using Monte Carlo simulation.

Teaching Assistant, University of Illinois Urbana-Champaign

AE 202, AE 443, AE 483

Jun 2021 - present

- ❖ Worked with 2-5 faculty and staff to teach lessons, hold office hours, and improve curriculum for courses in flight mechanics, systems engineering, and UAV controls.
- ❖ Assisted in developing code and instructions for a UAV lab based on C and Python.

SWE Intern, Collins Aerospace

May 2019 - Aug 2019

- ❖ Migrated an avionics data interpreter from 16-bit to 32-bit using C++ and Python.
- ❖ Automated builds in TortoiseSVN using Python, reducing need for manual integration.

HONORS AND AWARDS

Yee Fellowship

2021, 2024

Grainger College of Engineering, University of Illinois Urbana-Champaign

Aerospace Engineering Alumni Advisory Board Fellowship

2024

Department of Aerospace Engineering, University of Illinois Urbana-Champaign

Best Student Paper, 46th AAS GN&C Conference

2024

American Astronautical Society (AAS) Rocky Mountain Section

David and Catherine Thompson Space Technology Scholarship

2020

American Institute of Aeronautics and Astronautics (AIAA)